AWS Telecom Data Lake Architecture using AWS Glue and PySpark

This white paper describes a scalable and cost-effective AWS-based telecom data lake architecture designed for processing high-volume OSS (Operations Support System) data. The solution leverages AWS Glue, Amazon S3, PySpark, and Amazon Athena to build analytics-ready datasets for network performance analysis at cell and site level.

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1. Solution Architecture

The solution follows a Medallion-style data lake architecture implemented on Amazon S3. The data is organized into three layers: **Raw Layer:** Stores original OSS telecom data. **Curated Layer:** Stores cleaned and standardized datasets. **Processed Layer:** Stores aggregated KPI datasets for analytics and reporting. Data is partitioned by date and region/cluster to improve query performance and reduce data scan cost.

2. End-to-End Pipeline Flow

Step	Description
1	OSS telecom data ingestion into Amazon S3
2	Glue Crawlers update metadata in Glue Data Catalog
3	AWS Glue PySpark jobs perform transformation and KPI computation
4	Processed data stored in Parquet format on Amazon S3
5	Amazon Athena used for ad-hoc analytics and querying

3. Technologies Used

Amazon S3: Centralized scalable data lake storage. **AWS Glue:** Serverless ETL service for data processing. **PySpark:** Distributed processing for large-scale telecom datasets. **Parquet:** Columnar storage format for optimized analytics. **Amazon Athena:** Serverless querying on S3 data. **CloudWatch:** Monitoring and logging.

4. Performance Optimization Techniques

The following optimization strategies were implemented to improve ETL performance: Partitioning based on date and region. Broadcast joins for small lookup datasets. Conversion of CSV files into Parquet format. Partition pruning to reduce unnecessary data scan. Optimized Spark transformations and joins. These optimizations improved pipeline performance by approximately 30%.

5. Business Benefits

Scalable architecture for high-volume telecom data. Reduced storage and query cost. Improved query performance and analytics speed. Enabled faster identification of network issues such as drop rate and latency. Simplified serverless ETL operations.

6. Future Enhancements

Future enhancements include integration with Databricks Delta Lake, real-time streaming using Kafka/Kinesis, and implementation of CI/CD pipelines for automated deployment and monitoring.

7. Conclusion

This AWS-based telecom data lake architecture provides a scalable, reliable, and cost-efficient solution for processing and analyzing OSS telecom data. The combination of AWS Glue, PySpark, Amazon S3, and Athena enables efficient ETL processing and analytics-ready data generation for business insights.